

Getting Started with EEG Biomarker Critique

Step-by-step setup, data retrieval, validation, and contribution guide

Repository: [vaishnavighs205/eeg-biomarker-critique](https://github.com/vaishnavighs205/eeg-biomarker-critique)

OpenNeuro dataset: ds004504 v1.0.9

Updated August 2026

Purpose: Use this guide when setting up a new GitHub Codespace or local machine to work on the repository. It covers the software environment, OpenNeuro dataset, QC pipeline, epoch manifest, Git workflow, package management, and the scientific rules that must not be broken.

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1. Project Overview

This repository implements a reproducible biomedical engineering and computer science workflow for evaluating EEG biomarkers of Alzheimer's disease and determining whether robust biomarker information can be preserved using fewer scalp electrodes.

```

OpenNeuro ds004504
|
v
Dataset retrieval + metadata matching
|
v
Subject-level EEG quality control
|
v
30 s windows with 15 s overlap
|
v
DICE-style RBP + spectral coherence replication
|
v
Biomarker critique
|
+-- fixed vs individualized frequency bands
+-- relative vs absolute power
+-- periodic vs aperiodic spectral structure
+-- coherence vs wPLI / imaginary coherence
+-- AD vs FTD specificity
`-- MMSE association
|
v

```

```

Leakage-safe machine learning
|
v
Minimal-electrode optimization

```

Current verified dataset state: The QC pipeline has successfully loaded and validated all 88 subjects. The current epoch manifest contains 4,521 overlapping 30-second windows. These windows are repeated measurements from 88 people, not 4,521 independent subjects.

2. Repository Structure

```

eeg-biomarker-critique/
|-- README.md
|-- CONTRIBUTING.md
|-- config.yaml
|-- environment.yml
|-- requirements.txt
|-- pyproject.toml
|-- Makefile

|-- data/
|   |-- ds004504/          # local OpenNeuro dataset; never commit
|   |-- raw/
|   |-- derivatives/
|   `-- processed/

|-- docs/
|-- notebooks/

|-- results/
|   |-- figures/
|   |-- tables/
|   |-- models/
|   `-- logs/

|-- scripts/
|   |-- get_ds004504.sh
|   |-- test_subject.py
|   |-- run_qc.py
|   |-- build_epoch_manifest.py
|   |-- run_replication.py
|   `-- run_ml.py

|-- src/eeg_biomarkers/
`-- tests/

```

Where new work belongs

Purpose	Location
Reusable analysis code	src/eeg_biomarkers/
One-command pipeline entry points	scripts/
Exploratory analysis only	notebooks/
Small reproducible CSV outputs	results/tables/
Plots	results/figures/
Downloaded EEG data	data/ds004504/ - ignored by Git

Purpose	Location
Project instructions / methods	docs/

3. Prerequisites

- Git
- Conda / Miniconda / Mambaforge
- GitHub account with repository access
- GitHub CLI (gh) - recommended for authentication
- DataLad and git-annex for retrieving the EEG payload
- Python 3.11 inside the project environment

GitHub Codespaces: The repository works well in Codespaces. Conda may be installed but not initialized, and some Codespaces have no default Conda channels configured. The commands below explicitly handle both situations.

4. Clone the Repository

```
git clone https://github.com/vaishnavighs205/eeg-biomarker-critique.git
cd eeg-biomarker-critique
```

In GitHub Codespaces, the repository is normally already present. Confirm your location:

```
pwd
```

Expected Codespaces path:

```
/workspaces/eeg-biomarker-critique
```

5. Configure Git and GitHub Authentication

5.1 Configure commit identity

```
git config --global user.name "vaishnavighs205"
git config --global user.email "vaishnavighs205@gmail.com"

git config --global user.name
git config --global user.email
```

5.2 Authenticate GitHub pushes

GitHub does not accept your normal account password for HTTPS Git pushes. In Codespaces, use GitHub CLI authentication:

```
gh auth status
gh auth login
gh auth setup-git
```

Then test:

```
git push
```

6. Initialize Conda

If you see:

```
CondaError: Run 'conda init' before 'conda activate'
```

initialize Bash and reload the shell:

```
conda init bash
source ~/.bashrc
```

Confirm Conda can see environments:

```
conda info --envs
```

7. Create the Python Environment

7.1 Preferred method: environment.yml

The repository should contain an environment file similar to:

```
name: eeg_biomarkers

channels:
  - conda-forge

dependencies:
  - python=3.11
  - pip
  - git
  - git-annex
  - datalad
  - pip:
    - -r requirements.txt
    - -e .
```

Create the environment:

```
conda env create -f environment.yml
conda activate eeg_biomarkers
```

7.2 Fallback method if environment.yml is not yet available

```
conda create -n eeg_biomarkers -c conda-forge python=3.11 pip git datalad git-annex -y
conda activate eeg_biomarkers
pip install -r requirements.txt
pip install -e .
```

NoChannelsConfiguredError: If Conda reports that no channels are configured, either use `-c conda-forge` as shown above or configure it permanently with: `conda config --add channels conda-forge`

8. Install and Verify Packages

The repository currently depends on MNE, MNE-Connectivity, NumPy, SciPy, pandas, PyArrow, scikit-learn, statsmodels, specparam, antropy, Matplotlib, XGBoost, PyYAML, joblib, and pytest.

```
python --version
which python
```

Expected Python version:

```
Python 3.11.x
```

Verify core packages:

```
python - <<'PY'
import mne
import numpy
```

```
import pandas
import scipy
import sklearn

print("MNE:", mne.__version__)
print("NumPy:", numpy.__version__)
print("pandas:", pandas.__version__)
print("SciPy:", scipy.__version__)
print("scikit-learn:", sklearn.__version__)
PY
```

Verify dataset tools:

```
datalad --version
git annex version
```

9. config.yaml vs environment.yaml

File	Purpose	Examples
environment.yaml	Software environment	Python 3.11, DataLad, git-annex, pip dependencies
requirements.txt	Python package requirements	MNE, SciPy, pandas, scikit-learn, XGBoost
config.yaml	Scientific / analysis configuration	dataset path, 30 s epochs, frequency bands, CV folds, model settings

Do not replace config.yaml: config.yaml controls the analysis. environment.yaml controls the software environment. The repository needs both.

10. Retrieve OpenNeuro ds004504

The project uses OpenNeuro dataset ds004504 v1.0.9. The upstream GitHub mirror is a DataLad / git-annex repository, so a plain Git clone may give you file links without downloading the EEG payload.

10.1 Use the repository helper script

```
bash scripts/get_ds004504.sh
```

The expected destination is:

```
data/ds004504/
```

10.2 Manual retrieval

```
mkdir -p data
cd data

datalad clone https://github.com/OpenNeuroDatasets/ds004504.git
cd ds004504

datalad get derivatives/

cd ../../
```

11. Verify DataLad / git-annex Files

A .set file may appear as a symlink such as:

```
sub-001_task-eyesclosed_eeg.set -> ../../../../.git/annex/objects/...
```

That is normal for git-annex. Confirm the actual content is present:

```
cd data/ds004504

datalad get derivatives/sub-001/eeg/sub-001_task-eyesclosed_eeg.set

git annex whereis derivatives/sub-001/eeg/sub-001_task-eyesclosed_eeg.set

test -e derivatives/sub-001/eeg/sub-001_task-eyesclosed_eeg.set && echo "EEG data
available"

cd ../../
```

Never commit the dataset: data/ds004504/ is intentionally excluded from Git. The dataset is large and is managed independently through OpenNeuro/DataLad.

12. Load One EEG Subject

Before processing all 88 subjects, verify that MNE can read one cleaned derivative recording:

```
python scripts/test_subject.py
```

A successful run should report values similar to:

```
Successfully loaded EEG!
Sampling frequency: 500.0 Hz
Number of channels: 19
Duration: 599.798 seconds
```

The channel order in the file does not need to match the order written in documentation. Always select channels by their names rather than relying on fixed array indices.

13. Run the Full 88-Subject QC Pipeline

```
python scripts/run_qc.py
```

The QC pipeline checks items such as:

- EEG file availability
- 19 channels
- 500 Hz sampling frequency
- expected channel names
- recording duration
- NaNs
- flat channels
- metadata mapping for diagnosis, age, sex, and MMSE

Expected current result:

```
status
PASS    88
```

The output table is:

```
results/tables/dataset_qc.csv
```

14. Build the 30-Second Epoch Manifest

The replication workflow uses 30-second windows with 15-second overlap. Build the manifest with:

```
python scripts/build_epoch_manifest.py
```

Current verified manifest summary:

```
Subjects: 88
Total windows: 4521

Windows per diagnosis:
AD      1888
CN      1563
FTD     1070

Windows per subject:
min      19
median   52
max      84
```

Critical statistical rule: 4,521 windows do not mean $N=4,521$. The independent sample remains $N=88$ subjects. Windows from the same subject must never be divided across training and testing sets.

15. Understand Generated Result Files

The repository is configured so that the downloaded EEG data is ignored, while selected small CSV summary tables can be tracked. Generated models, logs, and most figures remain ignored unless intentionally added.

15.1 Check what Git ignores

```
git check-ignore -v results/tables/dataset_qc.csv
```

Current small tables you may intentionally track include:

- results/tables/dataset_qc.csv
- results/tables/epoch_manifest.csv
- future subject-level summary / model metric CSV files

Add them from the repository root:

```
cd /workspaces/eeg-biomarker-critique

git add .gitignore      results/tables/dataset_qc.csv      results/tables/epoch_manifest.csv

git commit -m "Track generated analysis tables"
git push
```

Run Git commands from the repository root: Doing Git operations from results/ or scripts/ is valid, but relative paths become easy to confuse. Prefer /workspaces/eeg-biomarker-critique.

16. Daily Development Workflow

```
cd /workspaces/eeg-biomarker-critique
conda activate eeg_biomarkers

python --version
git status
```

If you are using a local clone rather than Codespaces, replace the path with the location of your clone.

17. Pulling Updates and Updating Dependencies

```
git pull
```

Then refresh Python packages and editable installation:

```
pip install -r requirements.txt
pip install -e .
```

If environment.yml changed:

```
conda env update -f environment.yml --prune
```

18. Adding or Updating Python Packages

Do not install a package only into your own environment and leave it undocumented. If your code needs a new package:

1. Install it locally to test your work.
2. Add the dependency and minimum compatible version to requirements.txt.
3. If it is a Conda/system dependency, update environment.yml too.
4. Run the test suite.
5. Commit the dependency file together with the code that uses it.

```
pip install package-name
# then edit requirements.txt
pytest -q
```

19. Branching, Commits, and Pull Requests

19.1 Recommended contributor workflow

```
git pull

git checkout -b feature/descriptive-name

# make changes
pytest -q

git status
git add <files>
git commit -m "Describe the change"

git push -u origin feature/descriptive-name
```

Then open a Pull Request into main. Small, focused pull requests are easier to review and safer to merge.

19.2 Useful commit examples

- Add ds004504 loading and subject-level QC pipeline

- Add epoch manifest generation
- Implement relative band power extraction
- Add SCC replication
- Add individualized alpha frequency analysis
- Add wPLI connectivity analysis
- Add subject-level nested validation
- Add electrode selection pipeline

20. Scientific Validation Rules

Rule 1 - Subject is the independent sample: All windows belonging to one participant must remain within the same train/validation/test partition.

Rule 2 - Prevent leakage: Scaling, feature selection, hyperparameter tuning, threshold selection, and electrode selection must be learned only from training subjects.

Rule 3 - Reproduce before critique: First reproduce the DICE-style RBP and SCC workflow. Only then change fixed bands, spectral normalization, aperiodic modeling, or connectivity metrics.

Rule 4 - Do not over-interpret EEG: Scalp EEG does not directly measure synapses. Discuss spectral, aperiodic, and connectivity variables as electrophysiological markers associated with circuit or network dysfunction unless stronger evidence is available.

Rule 5 - Test disease specificity: A feature that differs between AD and controls but also differs similarly in FTD may be a general dementia/neurodegeneration marker rather than an AD-specific biomarker.

21. Troubleshooting

CondaError: Run conda init before conda activate

```
conda init bash
source ~/.bashrc
```

EnvironmentNameNotFound: eeg_biomarkers

```
conda info --envs
conda env create -f environment.yml
conda activate eeg_biomarkers
```

NoChannelsConfiguredError

```
conda config --add channels conda-forge
```

or create the environment with an explicit channel:

```
conda create -n eeg_biomarkers -c conda-forge python=3.11 -y
```

FileNotFoundError for data/ds004504

Check that you are running from the repository root and that the dataset exists:

```
pwd
ls data
ls -lh data/ds004504/derivatives/sub-001/eeg/
```

The .set file exists but MNE cannot open it

The git-annex content may not have been retrieved:

```
cd data/ds004504
datalad get derivatives/sub-001/eeg/sub-001_task-eyesclosed_eeg.set
cd ../../
```

git commit says Author identity unknown

```
git config --global user.name "vaishnavighs205"
git config --global user.email "vaishnavighs205@gmail.com"
```

git push asks for a password and fails

```
gh auth login
gh auth setup-git
git push
```

CSV files exist locally but do not appear on GitHub

```
git status
git check-ignore -v results/tables/dataset_qc.csv
```

If they are intended to be tracked, make sure .gitignore allows them, then add and commit them from the repository root.

22. First-Contribution Checklist

- ☐ Repository cloned or Codespace opened
- ☐ Git identity configured
- ☐ GitHub authentication works
- ☐ Conda initialized
- ☐ eeg_biomarkers environment created
- ☐ Python 3.11 verified
- ☐ requirements.txt installed
- ☐ editable package installation works
- ☐ DataLad and git-annex verified
- ☐ ds004504 derivatives downloaded
- ☐ sub-001 loads successfully in MNE
- ☐ 88/88 subjects pass QC
- ☐ epoch_manifest.csv generated
- ☐ pytest passes
- ☐ work is performed on a feature branch
- ☐ no EEG dataset files are staged for commit

[] no subject-level data leakage is introduced

Quick Reference

```
# Start work
cd /workspaces/eeg-biomarker-critique
conda activate eeg_biomarkers

# Update code
git pull

# Verify installation
python --version
pytest -q

# Retrieve dataset (first setup only)
bash scripts/get_ds004504.sh

# Validate dataset
python scripts/test_subject.py
python scripts/run_qc.py

# Build analysis windows
python scripts/build_epoch_manifest.py

# Check repository state
git status

# Commit work
git add <files>
git commit -m "Describe the change"
git push
```

Repository: <https://github.com/vaishnavighs205/eeg-biomarker-critique>

Dataset: <https://openneuro.org/datasets/ds004504/versions/1.0.9>